

3. Use Gaussian elimination.

b.

$$A = \begin{bmatrix} 2 & -2 & 4 \\ -1 & 1 & -2 \\ 3 & -3 & 6 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 2 \\ 1 & -1 & 2 \\ 1 & -1 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$A\mathbf{x} = \mathbf{0}$ if and only if

$$x_1 - x_2 + 2x_3 = 0$$

so then $\mathbf{x} = (x_1, x_2, x_3) = (x_2 - 2x_3, x_2, x_3) = x_2(1, 1, 0) + x_3(-2, 0, 1)$, or we can write

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 - 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

d.

$$\begin{aligned} A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 1 & 2 & 2 & 3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 1 & 3 \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &\rightsquigarrow \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \end{aligned}$$

$x_1 = x_4$, $x_2 = -x_4$, $x_3 = -x_4 \Rightarrow$ then $\mathbf{x} = (x_1, x_2, x_3, x_4) = (x_4, -x_4, -x_4, x_4) = x_4(1, -1, -1, 1)$.

4. Use Gaussian elimination to write $[A|\mathbf{b}]$ in reduced echelon form.

a.

$$\begin{aligned} [A|\mathbf{b}] &= \left[\begin{array}{ccc|c} 2 & 1 & -1 & 3 \\ 1 & 2 & 1 & 0 \\ -1 & 1 & 2 & -3 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 1 & -1 & 3 \\ -1 & 1 & 2 & -3 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -3 & 3 \\ 0 & 3 & 3 & -3 \end{array} \right] \\ &\rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & -3 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right], \end{aligned}$$

so then $x_1 - x_3 = 2$ and $x_2 + x_3 = -1$ which gives,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 + x_3 \\ -1 - x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

b.

$$\begin{aligned} [A|\mathbf{b}] &= \left[\begin{array}{cc|c} 2 & -1 & -4 \\ 2 & 1 & 0 \\ -1 & 1 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & -3 \\ 2 & -1 & -4 \\ 2 & 1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & -3 \\ 0 & 1 & 2 \\ 0 & 3 & 6 \end{array} \right] \\ &\rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 3 & 6 \end{array} \right], \end{aligned}$$

so then $x_1 = -1$ and $x_2 = 2$ which gives the solution,

$$\mathbf{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

d.

$$\begin{aligned}
[A|\mathbf{b}] &= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 2 & 1 & 3 & -1 \\ 1 & 1 & 2 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 2 & -1 & 1 & 1 \\ 2 & 1 & 3 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -3 & -3 & 3 \\ 0 & -1 & -1 & 1 \end{array} \right] \\
&= \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -3 & -3 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right],
\end{aligned}$$

So then $x_1 = -x_3$ and $x_2 = -1 - x_3$ which gives the solution,

$$\mathbf{x} = \begin{bmatrix} -x_3 \\ -1 - x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$

5. The hint suggests rewriting, $A\mathbf{x} = \mathbf{x}$ as $B\mathbf{x} = \mathbf{0}$ for some matrix B . $A\mathbf{x} = \mathbf{x} \Rightarrow A\mathbf{x} - 1\mathbf{x} = \mathbf{0}$. For “1” we can use the matrix

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

i.e., the matrix with a diagonal of 1's and 0's elsewhere.

b.

$$A - I = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 2 \\ -2 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ -2 & 0 & 2 \\ -2 & 0 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so then $x_1 = x_3$ which gives,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$